

Hybrid Arguments and CCA Security

CS 171: Cryptography

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September 15, 2026

Overview

- 1 Plan for Today
- 2 Recap: CPA-Security
- 3 Mult-Security $\nRightarrow$ CPA-Security
- 4 Hybrid Arguments: CPA-Security $\Rightarrow$ Mult-Security
- 5 CCA Security
- 6 Summary

Plan for Today

- **Left over from last time:** mult-security vs. CPA-security.
 - A scheme that is mult-secure but *not* CPA-secure.
 - The other direction, CPA-security $\Rightarrow$ mult-security — a new proof technique: **hybrid arguments**.
 - Together: CPA-security is *strictly* stronger than mult-security.
- **Chosen-ciphertext attacks** and CCA-security — stronger still.

Recap: CPA-Security

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}(n)$

- 1 $k \leftarrow \text{Gen}(1^n)$. $\mathcal{A}$ is given 1^n and **oracle access** to $\text{Enc}_k(\cdot)$.
- 2 $\mathcal{A}^{\text{Enc}_k(\cdot)}$ outputs $m_0, m_1 \in \{0, 1\}^*$ with $|m_0| = |m_1|$.
- 3 $b \leftarrow \{0, 1\}$, $c^* \leftarrow \text{Enc}_k(m_b)$.
- 4 c^* is given to $\mathcal{A}^{\text{Enc}_k(\cdot)}$.
- 5 $\mathcal{A}^{\text{Enc}_k(\cdot)}$ outputs a guess b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

Definition 1 (CPA-Security)

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is *CPA-secure* if for all PPT $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Recap: CPA-Secure Encryption from a PRF

Construction

Let F be a PRF, $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$. Define $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$:

- $\text{Gen}(1^n)$: choose uniform $k \leftarrow \{0, 1\}^n$ and output it as the key.
- $\text{Enc}_k(m)$ (for $m \in \{0, 1\}^n$): sample uniform $r \leftarrow \{0, 1\}^n$ and output the ciphertext

$$c := \langle r, F_k(r) \oplus m \rangle.$$

- $\text{Dec}_k(c)$: parse $c = \langle r, s \rangle$ and output $m := F_k(r) \oplus s$.

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- $\text{Dec}_k(c)$: parse $c = \langle r, s \rangle$ and output $m := F_k(r) \oplus s$.
- **CPA-secure** if F is a PRF (proved last time, by reduction to the PRF game).
- Encryption is **randomized** — necessary: no deterministic (stateless) scheme is CPA-secure.
- Encrypts long messages block by block; ciphertext is twice the message length.

Recap: Mult-Security (Many Messages, No Oracle)

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$

- 1 $\mathcal{A}$ is given the security parameter 1^n .
- 2 For each $i \in \{1, \dots, t\}$, $\mathcal{A}$ outputs $m_{0,i}, m_{1,i} \in \{0, 1\}^*$ with $|m_{0,i}| = |m_{1,i}|$.
- 3 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$, $c_i \leftarrow \text{Enc}_k(m_{b,i})$ for $i = 1, \dots, t$.
- 4 $c_1, \dots, c_t$ is given to $\mathcal{A}$.
- 5 $\mathcal{A}$ outputs a guess b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

Definition 2 (Mult-Security)

Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has **indistinguishable multiple encryptions** in the presence of an eavesdropper (is *mult-secure*) if for all PPT $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

CPA-Security Is *Strictly* Stronger Than Mult-Security

- **How can we prove this?**

- Construct an encryption scheme Π that is **mult-secure** but **not CPA-secure**.
- *Simplify the problem*: assume Φ is mult-secure *and* CPA-secure; we weaken Φ to get Π so that Π is mult-secure but not CPA-secure.

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The (weird) scheme Π

Given $\Phi = (\text{Gen}, \text{Enc}, \text{Dec})$, set $\Pi = (\text{Gen}', \text{Enc}', \text{Dec}')$:

- $\text{Gen}'(1^n)$: set $k' := (k, m^*)$, where $k, m^* \leftarrow \text{Gen}(1^n)$.
- $\text{Enc}'_{k'}(m)$: if $m = m^*$ then output m^* . Otherwise, output $\text{Enc}_k(m) \parallel m^*$.
- $\text{Dec}'_{k'}(c)$: define naturally!

Need to prove that (1) Π is **mult-secure**, but (2) Π is **not CPA-secure**!

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Need to prove that (1) Π is **mult-secure**, but (2) Π is **not CPA-secure**!

Why are we looking at this weird scheme?

(1) Π Is Mult-Secure: The Reduction $\mathcal{B}$

Mult challenger for Φ :
 $k \leftarrow \text{Gen}(1^n)$

Reduction / mult adversary $\mathcal{B}$ for Φ

Mult adversary $\mathcal{A}$

Given / To prove

Given: $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$ for non-negligible ϵ .

To prove: $\Pr[\text{PrivK}_{\mathcal{B}, \Phi}^{\text{mult}}(n) = 1] \geq \frac{1}{2} + \delta(n)$ for non-negligible δ .

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$\vec{m}_0, \vec{m}_1$

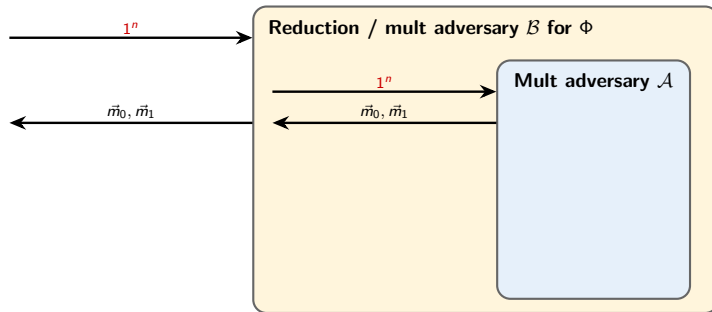
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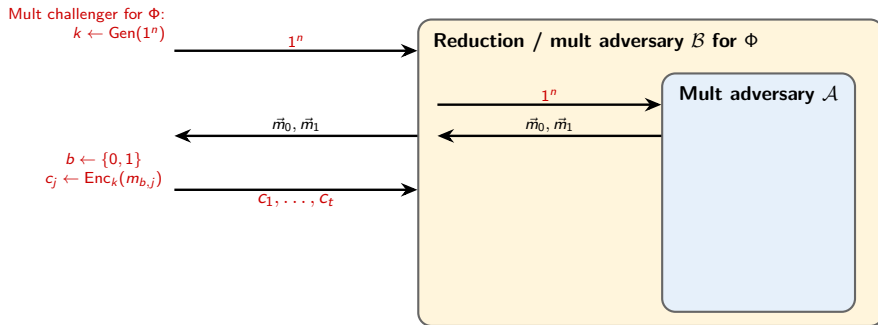


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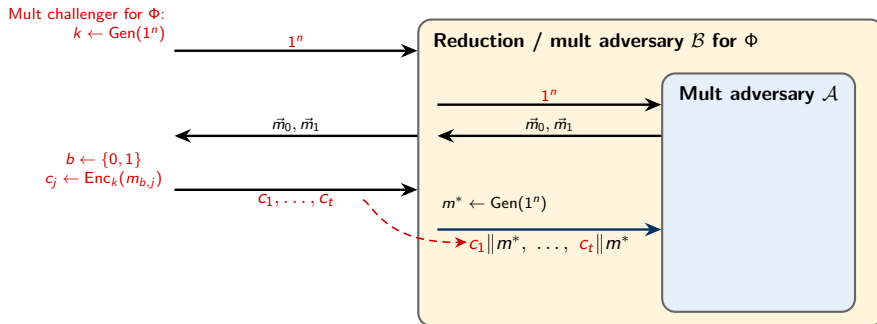


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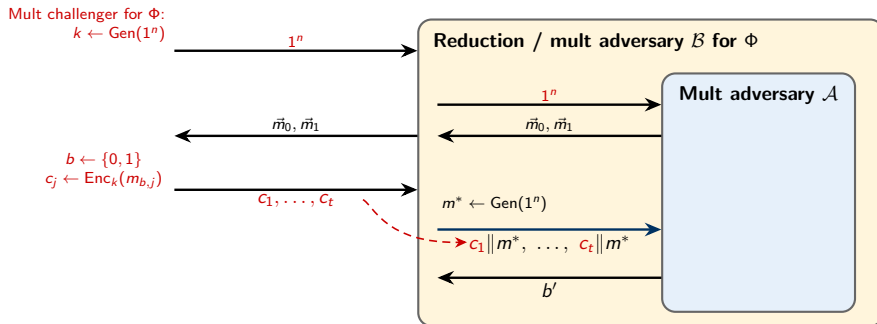


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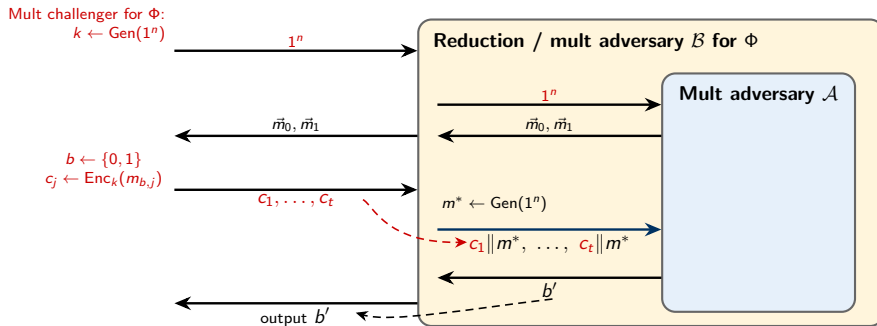


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(1) Π Is Mult-Secure: The Argument

View of $\mathcal{A}$

- In $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$, $\mathcal{A}$ receives $\text{Enc}'_{k'}(m_{b,j}) = \text{Enc}_k(m_{b,j}) \parallel m^*$ for every j — *unless* some $m_{b,j} = m^*$.
- $\mathcal{B}$ shows $\mathcal{A}$ exactly this, with a fresh m^* ; the only difference is that $\mathcal{B}$'s encryption is **incorrect** if some $m_{b,j} = m^*$ (it sends $\text{Enc}_k(m^*) \parallel m^*$ where Π would send m^*).

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The bad event is rare

- $\mathcal{A}$ chooses its messages **up front**, with **no encryption oracle**, so it has *no information* about m^* .
- m^* is a fresh key with at least n uniform bits, so $\Pr[\exists j : m_{b,j} = m^*] \leq \frac{2t}{2^n}$ — negligible.

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Conclusion

$\Pr[\text{PrivK}_{\mathcal{B},\Phi}^{\text{mult}}(n) = 1] \geq \Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult}}(n) = 1] - \frac{2t}{2^n} \geq \frac{1}{2} + \epsilon(n) - \frac{2t}{2^n}$, a non-negligible advantage — contradicting mult-security of Φ . Hence Π is mult-secure. $\square$

(2) Π Is Not CPA-Secure

CPA attack $\mathcal{A}$

- 1 Query $\text{Enc}'_{k'}(\cdot)$ on input 0^n , and let $c \parallel m^*$ be the received ciphertext. ($\mathcal{A}$ now knows m^* !)
- 2 Submit challenge messages $m_0 = m^*$ and $m_1 = 0^n$.
- 3 Output 0 if $c^* = m^*$, and 1 otherwise.

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 - ② Submit challenge messages $m_0 = m^*$ and $m_1 = 0^n$.
 - ③ Output 0 if $c^* = m^*$, and 1 otherwise.
- If $b = 0$: $c^* = \text{Enc}'_{k'}(m^*) = m^*$, so $\mathcal{A}$ outputs 0.
 - If $b = 1$: $c^* = \text{Enc}_k(0^n) \parallel m^* \neq m^*$, so $\mathcal{A}$ outputs 1.
 - $\mathcal{A}$ wins with probability 1 (whenever $m^* \neq 0^n$).

The Other Direction: CPA-Security $\Rightarrow$ Mult-Security?

- We now know: mult-secure $\not\Rightarrow$ CPA-secure. Is CPA-security at least *as strong as* mult-security?
- Intuitively yes: the CPA adversary is *more* powerful (it has an encryption oracle) and can pick its challenge *after* seeing ciphertexts.

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But the reduction is not immediate

The CPA challenger encrypts **one** challenge pair, while the mult adversary expects t challenge ciphertexts. Bridging this gap needs a new tool: **hybrid arguments**.

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Conclusion (after the proof)

CPA-security is **strictly stronger** than mult-security: CPA-secure $\Rightarrow$ mult-secure, but mult-secure $\nRightarrow$ CPA-secure.

Theorem 3

*If Π is **CPA-secure**, then Π is **mult-secure**.*

CPA-Security $\Rightarrow$ Mult-Security

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Proof by reduction

Given a PPT adversary $\mathcal{A}$ “breaking” mult-security of Π , construct a PPT adversary $\mathcal{B}$ “breaking” CPA-security of Π .

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New proof strategy: hybrid arguments

$\mathcal{A}$ expects t challenge ciphertexts, but $\mathcal{B}$'s challenger produces only *one*. We bridge the gap with a sequence of **hybrid** experiments, changing one ciphertext at a time.

The Games Side by Side

$\text{PrivK}_{\mathcal{B}, \Pi}^{\text{CPA}}(n)$: what $\mathcal{B}$ plays

- 1 $k \leftarrow \text{Gen}(1^n)$; $\mathcal{B}$ is given 1^n and oracle access to $\text{Enc}_k(\cdot)$.
- 2 $\mathcal{B}^{\text{Enc}_k(\cdot)}$ outputs m_0^*, m_1^* with $|m_0^*| = |m_1^*|$.
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$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$: what $\mathcal{A}$ expects

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- 2 $\mathcal{A}$ outputs $(m_{0,i}, m_{1,i})$ with $|m_{0,i}| = |m_{1,i}|$, for $i = 1, \dots, t$.
- 3 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$; $c_i \leftarrow \text{Enc}_k(m_{b,i})$ for all i .
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- The hybrids are *thought experiments*: nobody runs them; they exist only in the proof.

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- 3 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$;
 $c_i \leftarrow \text{Enc}_k(0^{|m_{b,i}|})$ for $i \leq j$,
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- 4 $c_1, \dots, c_t$ is given to $\mathcal{A}$.
- 5 $\mathcal{A}$ outputs b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

- The hybrids are *thought experiments*: nobody runs them; they exist only in the proof.
- $\text{PrivK}^{\text{mult}, 0}$ is exactly $\text{PrivK}^{\text{mult}}$, so $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, 0}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$.

The Games Side by Side

$\text{PrivK}_{\mathcal{B}, \Pi}^{\text{CPA}}(n)$: what $\mathcal{B}$ plays

- 1 $k \leftarrow \text{Gen}(1^n)$; $\mathcal{B}$ is given 1^n and oracle access to $\text{Enc}_k(\cdot)$.
- 2 $\mathcal{B}^{\text{Enc}_k(\cdot)}$ outputs m_0^*, m_1^* with $|m_0^*| = |m_1^*|$.
- 3 $\hat{b} \leftarrow \{0, 1\}$, $c^* \leftarrow \text{Enc}_k(m_b^*)$.
- 4 c^* is given to $\mathcal{B}^{\text{Enc}_k(\cdot)}$.
- 5 $\mathcal{B}^{\text{Enc}_k(\cdot)}$ outputs $\hat{b}'$.
- 6 Output 1 if $\hat{b} = \hat{b}'$, and 0 otherwise.

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$: what $\mathcal{A}$ expects

- 1 $\mathcal{A}$ is given 1^n .
- 2 $\mathcal{A}$ outputs $(m_{0,i}, m_{1,i})$ with $|m_{0,i}| = |m_{1,i}|$, for $i = 1, \dots, t$.
- 3 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$; $c_i \leftarrow \text{Enc}_k(m_{b,i})$ for all i .
- 4 $c_1, \dots, c_t$ is given to $\mathcal{A}$.
- 5 $\mathcal{A}$ outputs b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, j}(n)$: hybrid j

- 1 $\mathcal{A}$ is given 1^n .
- 2 $\mathcal{A}$ outputs $(m_{0,i}, m_{1,i})$ with $|m_{0,i}| = |m_{1,i}|$, for $i = 1, \dots, t$.
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- 5 $\mathcal{A}$ outputs b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

- The hybrids are *thought experiments*: nobody runs them; they exist only in the proof.
- $\text{PrivK}^{\text{mult}, 0}$ is exactly $\text{PrivK}^{\text{mult}}$, so $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, 0}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$.
- In $\text{PrivK}^{\text{mult}, t}$ every c_i encrypts zeros, so $\mathcal{A}$'s view is independent of b :
 $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, t}(n) = 1] = \frac{1}{2}$.

Step 1: The Chain of Hybrids

$$\begin{aligned}c_1 &= \text{Enc}_k(m_{b,1}) \\c_2 &= \text{Enc}_k(m_{b,2}) \\&\vdots \\c_t &= \text{Enc}_k(m_{b,t})\end{aligned}$$

$$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, 0}(n)$$

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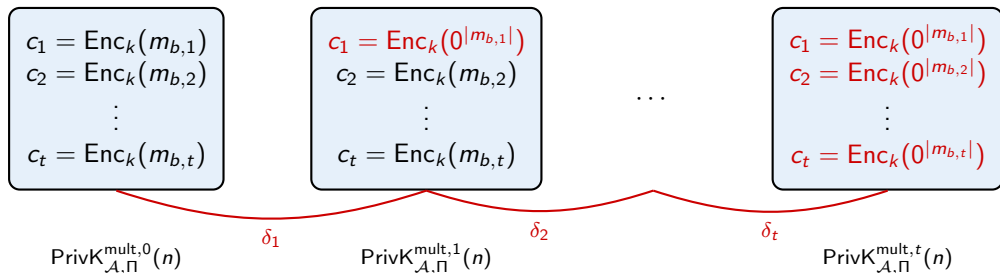
$$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, 1}(n)$$

...

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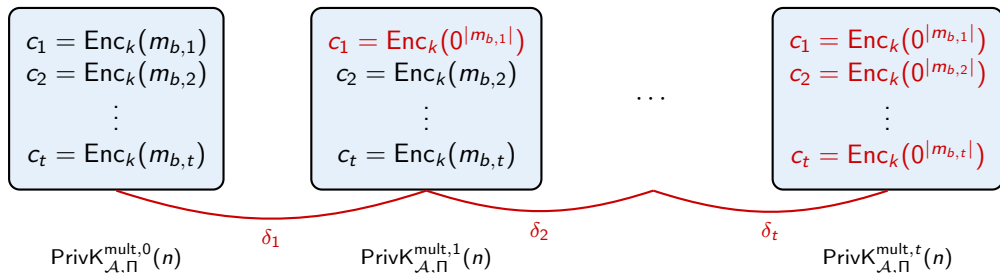
$$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, t}(n)$$

Step 1: The Chain of Hybrids



- Define $\delta_i := \left| \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i-1}(n) = 1] - \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i}(n) = 1] \right|$.

Step 1: The Chain of Hybrids



- Define $\delta_i := \left| \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i-1}(n) = 1] - \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i}(n) = 1] \right|$.
- From the previous slide, $\Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},0}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$ and $\Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},t}(n) = 1] = \frac{1}{2}$, so

$$\left| \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},0}(n) = 1] - \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},t}(n) = 1] \right| \geq \epsilon(n).$$

Step 2: Some Adjacent Pair Has a Large Gap

- Write $p_j := \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, j}(n) = 1]$. Then

$$\epsilon(n) \leq |p_0 - p_t|$$

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$$\epsilon(n) \leq |p_0 - p_t| = \left| \sum_{i=1}^t (p_{i-1} - p_i) \right| \quad (\text{telescoping: the middle terms cancel})$$

Key observation

$\text{PrivK}^{\text{mult}, i-1}$ and $\text{PrivK}^{\text{mult}, i}$ differ in *exactly one* ciphertext: c_i encrypts either $m_{b,i}$ or $0^{|m_{b,i}|}$.
That is exactly one CPA challenge!

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- A sum of t terms is $\geq \epsilon(n)$, so **at least one** $\delta_i \geq \epsilon(n)/t$ — non-negligible, since t is polynomial.

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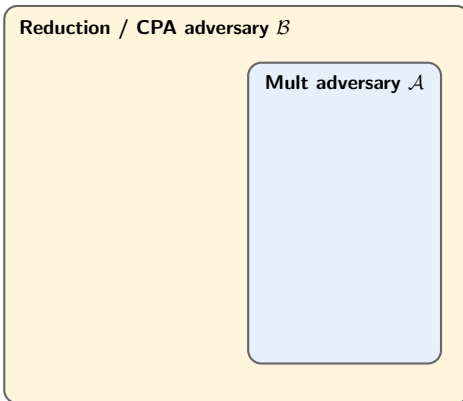
- A sum of t terms is $\geq \epsilon(n)$, so **at least one** $\delta_i \geq \epsilon(n)/t$ — non-negligible, since t is polynomial.
- Use this $\mathcal{A}$ — which distinguishes $\text{PrivK}^{\text{mult}, i-1}$ from $\text{PrivK}^{\text{mult}, i}$ — to **break CPA-security**.

Key observation

$\text{PrivK}^{\text{mult}, i-1}$ and $\text{PrivK}^{\text{mult}, i}$ differ in *exactly one* ciphertext: c_i encrypts either $m_{b,i}$ or $0^{|m_{b,i}|}$. That is exactly one CPA challenge!

Step 3: The Reduction $\mathcal{B}$ (for a Fixed i)

CPA challenger:
 $k \leftarrow \text{Gen}(1^n)$



Given / To prove

Given: $\delta_i(n)$ is non-negligible (WLOG the signed difference is $+\delta_i(n)$; else $\mathcal{B}$ flips its output).

To prove: $\Pr[\text{PrivK}_{\mathcal{B}, \Pi}^{\text{CPA}}(n) = 1] \geq \frac{1}{2} + \delta(n)$ for non-negligible δ .

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 $k \leftarrow \text{Gen}(1^n)$

1^n

Reduction / CPA adversary $\mathcal{B}$

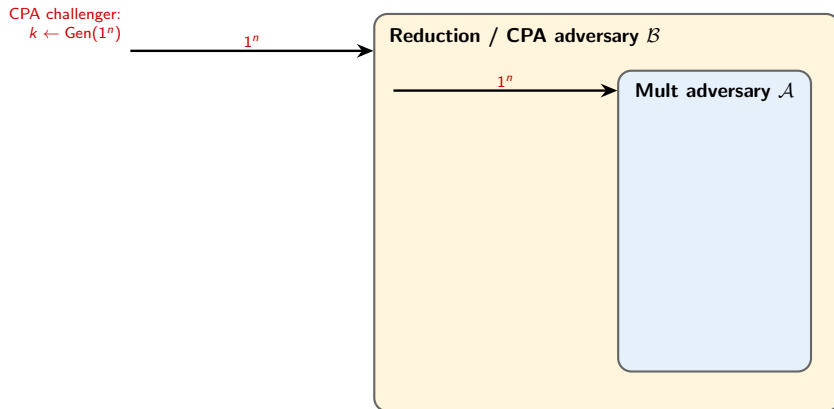
Mult adversary $\mathcal{A}$

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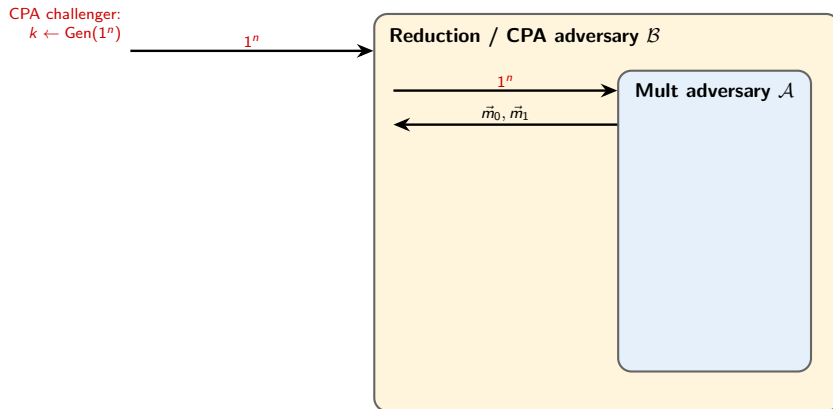


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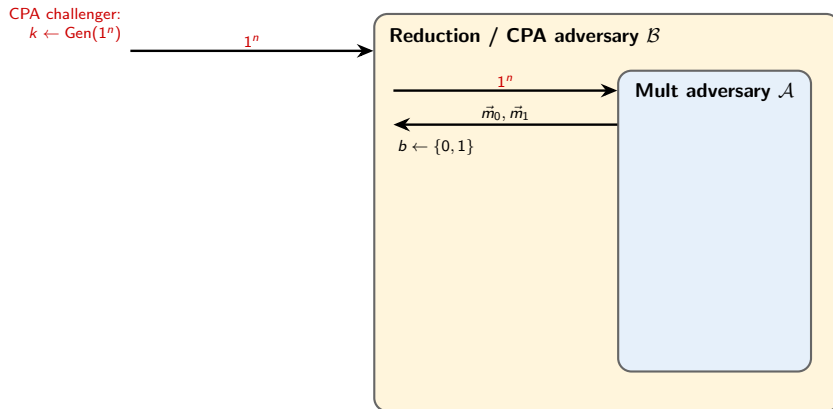


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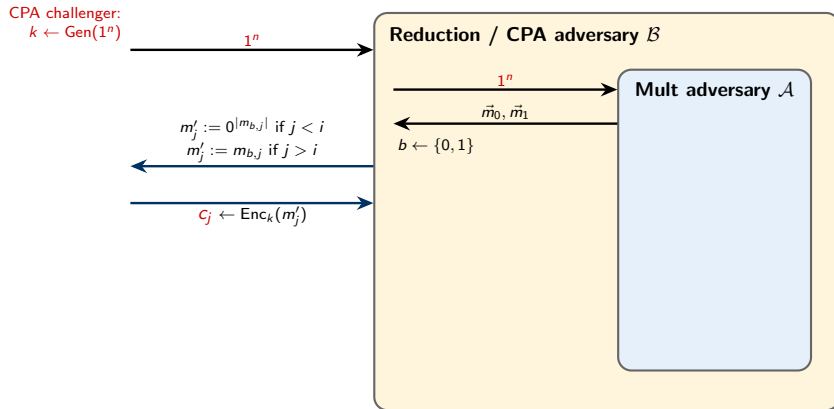


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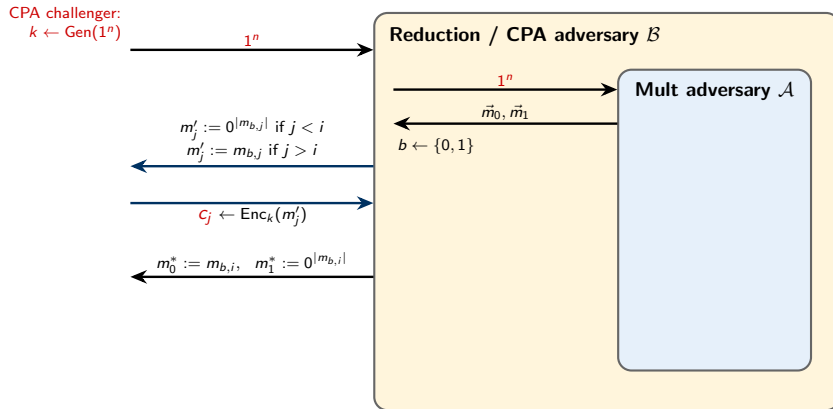


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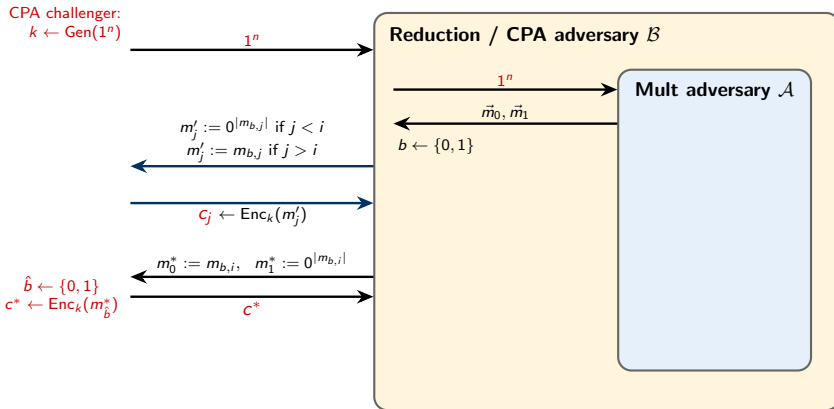


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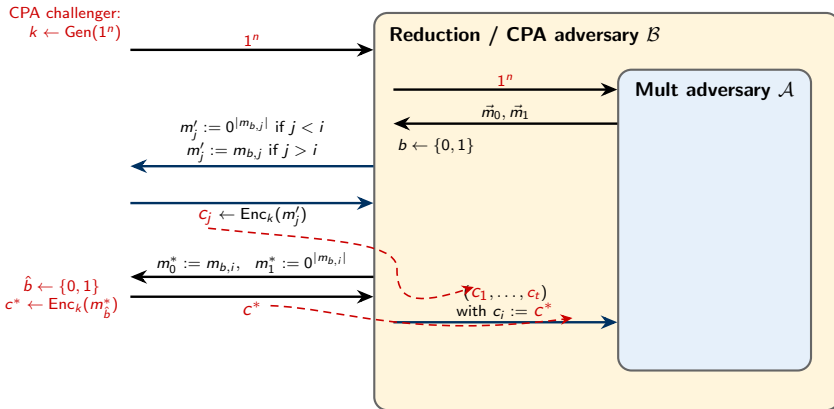


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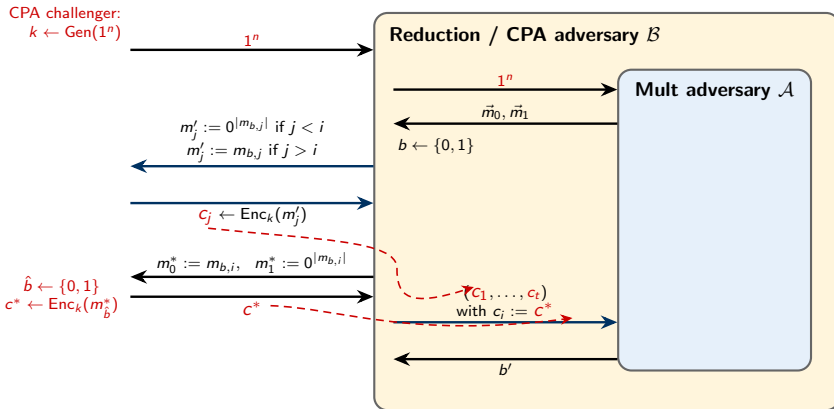


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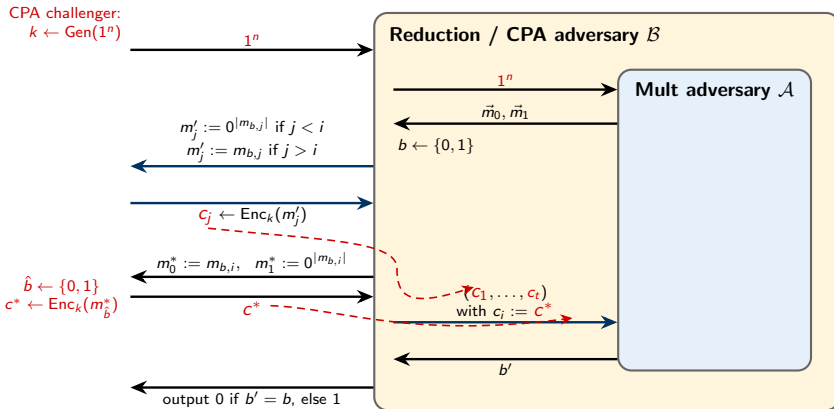


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Step 3: The Reduction $\mathcal{B}$ (in Words)

$\mathcal{B}$, given 1^n and oracle access to $\text{Enc}_k(\cdot)$

$\mathcal{B}$ runs $\mathcal{A}(1^n)$ and plays the mult challenger for it, using its oracle and its own challenge in place of Enc_k :

- ① Receive $\vec{m}_0, \vec{m}_1$ from $\mathcal{A}$. Sample $b \leftarrow \{0, 1\}$.
- ② For $j < i$: query $\text{Enc}_k(\cdot)$ on $0^{|m_{b,j}|}$ to get c_j . For $j > i$: query $\text{Enc}_k(\cdot)$ on $m_{b,j}$ to get c_j .
- ③ Submit the challenge $m_0^* := m_{b,i}$, $m_1^* := 0^{|m_{b,i}|}$; receive c^* and set $c_i := c^*$.
- ④ Give $(c_1, \dots, c_t)$ to $\mathcal{A}$; receive b' .
- ⑤ Output $\hat{b}' := 0$ (" c^* encrypts $m_{b,i}$ ") if $b' = b$, and $\hat{b}' := 1$ (" c^* encrypts $0^{|m_{b,i}|}$ ") otherwise.

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- $\mathcal{B}$ never needs k — the encryption oracle produces every c_j with $j \neq i$, and the challenger produces c_i .
- $\mathcal{B}$ is PPT, given $\mathcal{A}$ is PPT (t is polynomial).

Step 4: Two Cases

Case $\hat{b} = 0$ (c^* encrypts $m_{b,i}$)

$\mathcal{A}$'s view is exactly $\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i-1}(n)$.

Case $\hat{b} = 1$ (c^* encrypts $0^{|m_{b,i}|}$)

$\mathcal{A}$'s view is exactly $\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i}(n)$.

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Positions $j < i$ hold encryptions of $0^{|m_{b,j}|}$,
position i holds an encryption of $m_{b,i}$,
positions $j > i$ hold encryptions of $m_{b,j}$ — all
under the same uniform k , with fresh
randomness. That is hybrid $i - 1$.

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$\mathcal{A}$'s view is exactly $\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i}(n)$.

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Case $\hat{b} = 1$ (c^* encrypts $0^{|m_{b,i}|}$)

$\mathcal{A}$'s view is exactly $\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult},i}(n)$.

Now positions $1, \dots, i$ all hold encryptions of zeros and positions $i + 1, \dots, t$ hold the real messages. That is hybrid i .

Step 4: Putting It Together

Success probability of $\mathcal{B}$

$\mathcal{B}$ wins iff $b' = b$ when $\hat{b} = 0$ (view: hybrid $i - 1$), and $b' \neq b$ when $\hat{b} = 1$ (view: hybrid i):
 $\Pr[\text{PrivK}_{\mathcal{B}, \Pi}^{\text{CPA}}(n) = 1]$

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Success probability of $\mathcal{B}$

$\mathcal{B}$ wins iff $b' = b$ when $\hat{b} = 0$ (view: hybrid $i - 1$), and $b' \neq b$ when $\hat{b} = 1$ (view: hybrid i):

$$\begin{aligned}\Pr[\text{PrivK}_{\mathcal{B}, \Pi}^{\text{CPA}}(n) = 1] &= \frac{1}{2} \Pr[b' = b \mid \hat{b} = 0] + \frac{1}{2} \Pr[b' \neq b \mid \hat{b} = 1] \\ &= \frac{1}{2} \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i-1}(n) = 1] + \frac{1}{2} \left(1 - \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i}(n) = 1]\right) \\ &= \frac{1}{2} + \frac{1}{2} \left(\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i-1}(n) = 1] - \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}, i}(n) = 1]\right) = \frac{1}{2} + \frac{\delta_i(n)}{2}.\end{aligned}$$

Since $\delta_i(n) \geq \epsilon(n)/t$ is non-negligible, so is $\delta_i(n)/2$, contradicting that Π is CPA-secure.
Hence CPA-security $\Rightarrow$ mult-security. □

Chosen-Ciphertext Attacks

Chosen-ciphertext attack (CCA)

We strengthen the attacker's power once more: in addition to an **encryption** oracle, the adversary also gets a **decryption** oracle, which it may query on any ciphertext other than the challenge ciphertext.

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- Models settings where the attacker can observe how a system *reacts* to ciphertexts it submits (error messages, timing, responses, ...).
- Realistic: padding-oracle attacks on real protocols are exactly of this form.
- As we will see, it is **strictly stronger** than CPA-security.

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CCA}}(n)$

- ① $k \leftarrow \text{Gen}(1^n)$. $\mathcal{A}$ is given 1^n and **oracle access** to $\text{Enc}_k(\cdot)$ and $\text{Dec}_k(\cdot)$.
- ② $\mathcal{A}^{\text{Enc}_k(\cdot), \text{Dec}_k(\cdot)}$ outputs $m_0, m_1 \in \{0, 1\}^*$ with $|m_0| = |m_1|$.
- ③ $b \leftarrow \{0, 1\}$, $c^* \leftarrow \text{Enc}_k(m_b)$.
- ④ c^* is given to $\mathcal{A}^{\text{Enc}_k(\cdot), \text{Dec}_k(\cdot)}$, which may **not** query $\text{Dec}_k(\cdot)$ on c^* .
- ⑤ $\mathcal{A}^{\text{Enc}_k(\cdot), \text{Dec}_k(\cdot)}$ outputs a guess b' .
- ⑥ Output 1 if $b = b'$, and 0 otherwise.

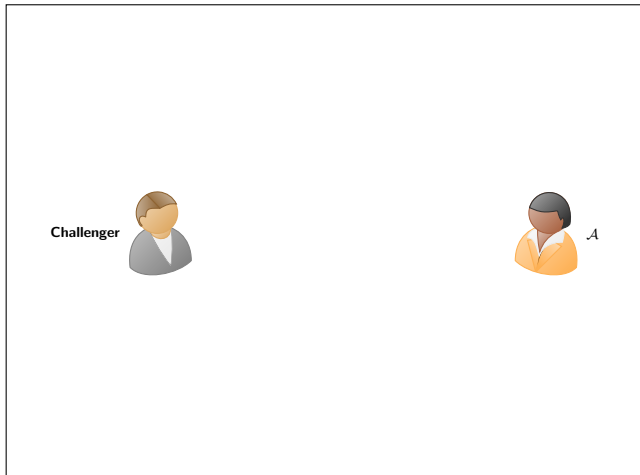
Definition 4 (CCA-Security)

Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has **indistinguishable encryptions under a chosen-ciphertext attack** (is *CCA-secure*) if for all PPT $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CCA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

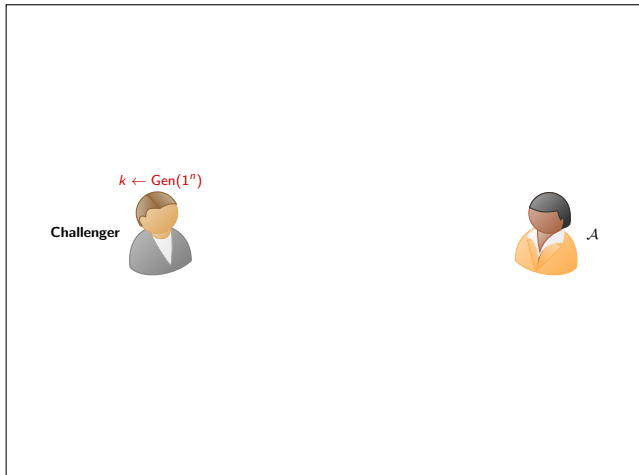
$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CCA}}$: Challenger vs. Adversary

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CCA}}(n)$

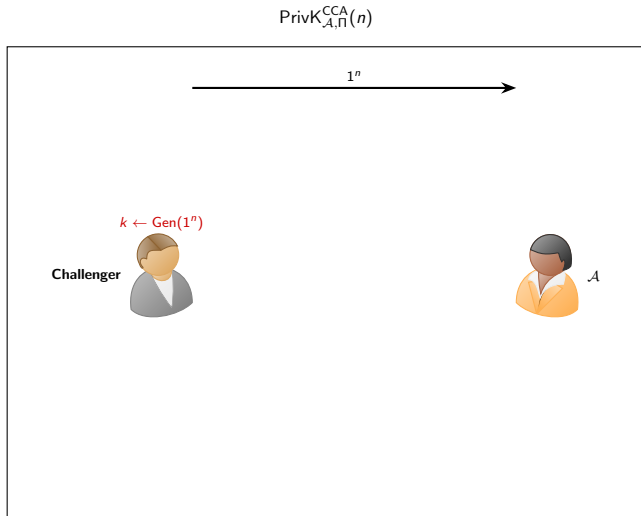


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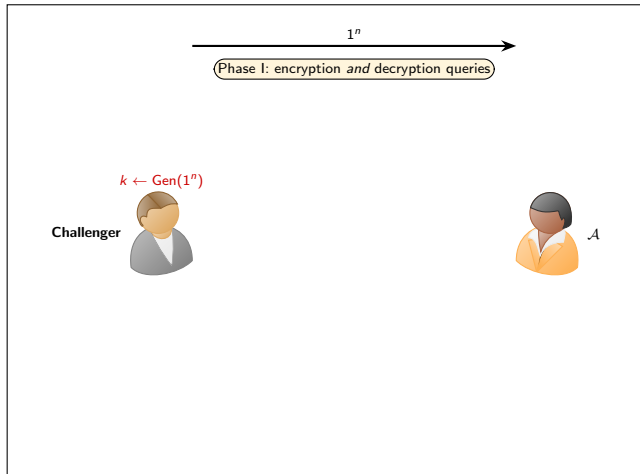


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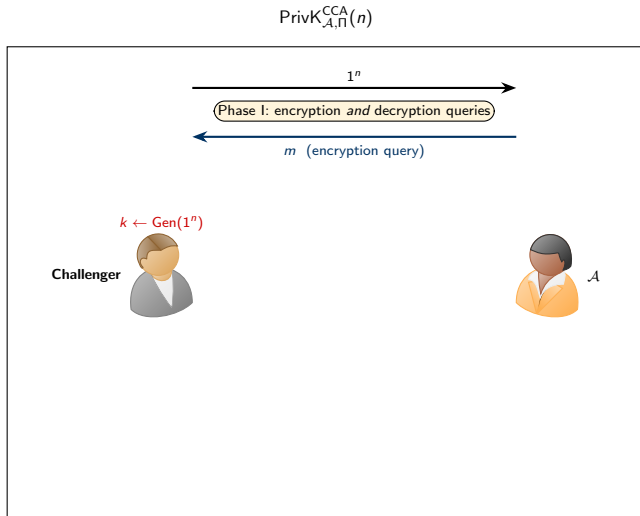


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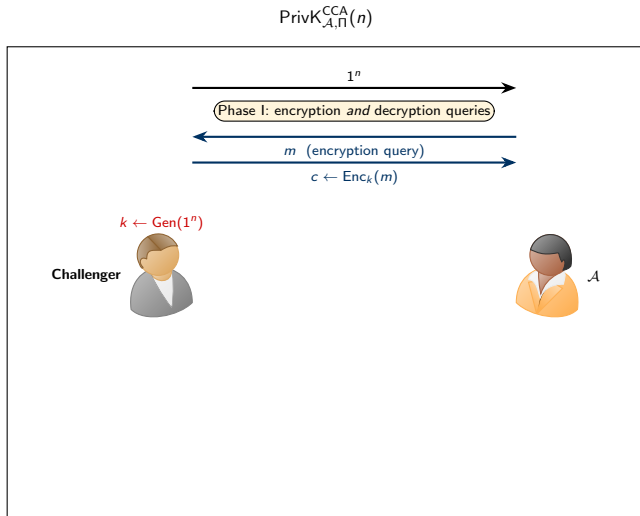
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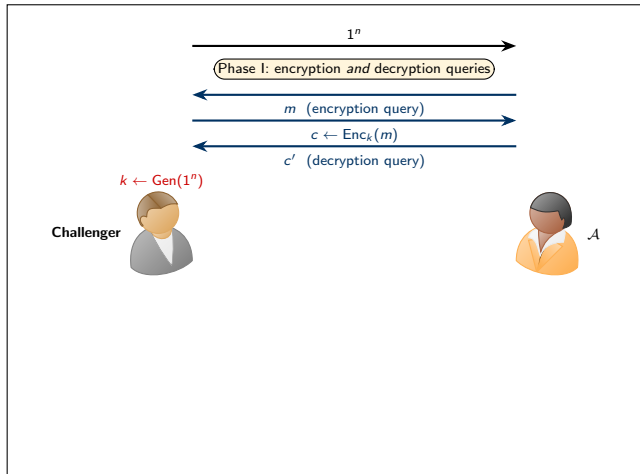


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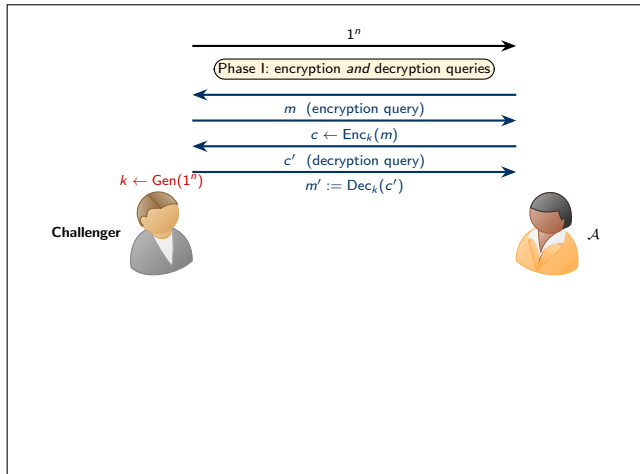
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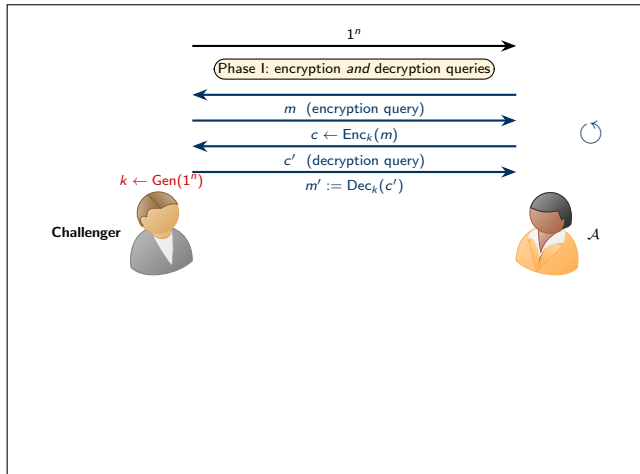
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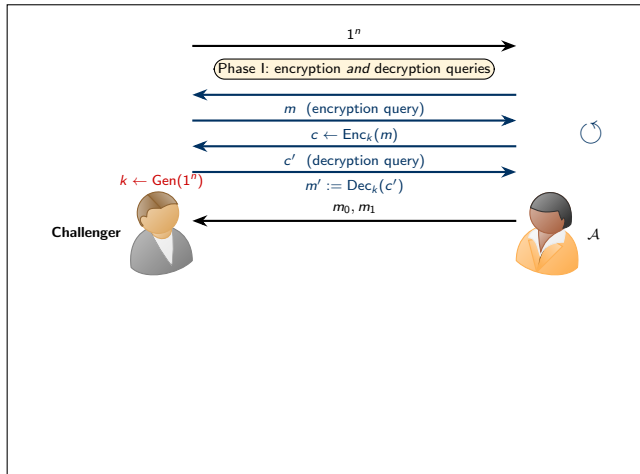
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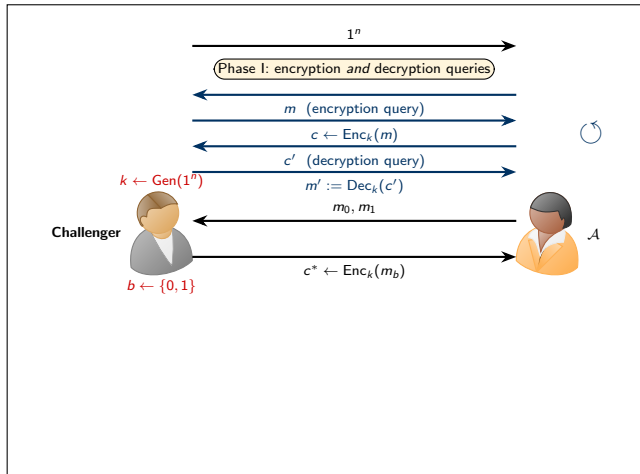
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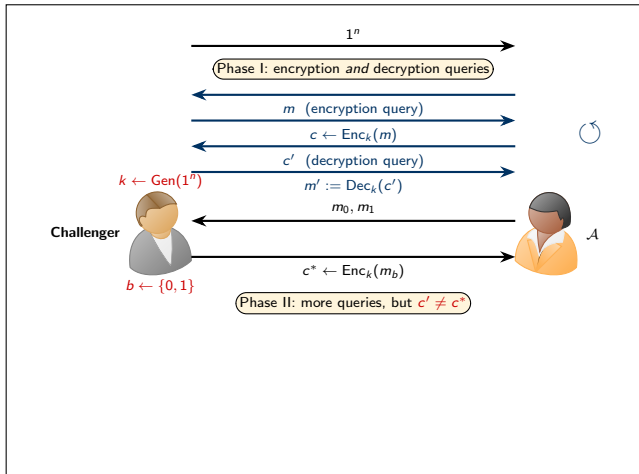
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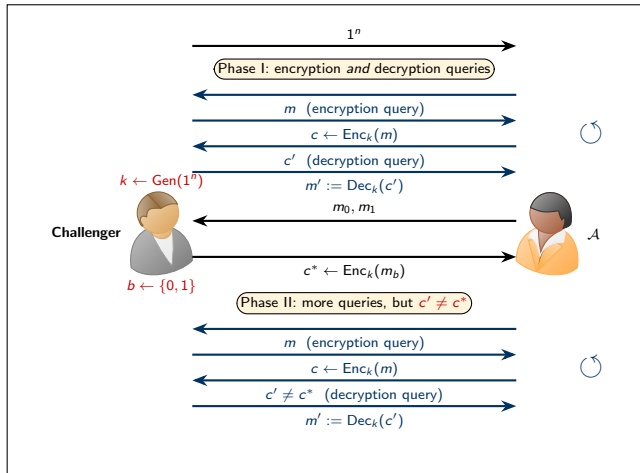
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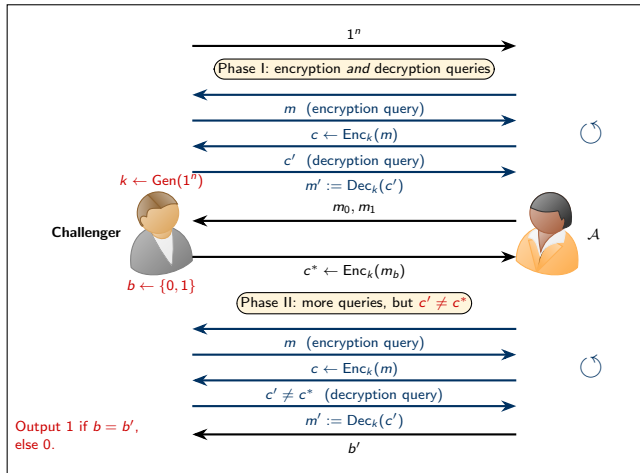
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Is Π CCA-Secure?

Construction (recall)

Let F be a PRF, $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$.

- $\text{Gen}(1^n)$: choose uniform $k \leftarrow \{0, 1\}^n$ and output it as the key.
- $\text{Enc}_k(m)$ (for $m \in \{0, 1\}^n$): sample uniform $r \leftarrow \{0, 1\}^n$ and output the ciphertext

$$c := \langle r, F_k(r) \oplus m \rangle.$$

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- This scheme is **CPA-secure** (last lecture). Question: is it also **CCA-secure**?
- **Hint**: what happens to the plaintext if we flip one bit of s ?

No! A CCA Attack on Π

No, here is an attacker $\mathcal{A}$ (recall $\text{Dec}_k(\langle r, s \rangle) = F_k(r) \oplus s$). Let $\Delta := 0^{n-1}1$.

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- ① $\mathcal{A}$ submits challenge messages $m_0 = 0^n$ and $m_1 = 1^n$.
- ② Challenger gives $c^* = \langle r^*, s^* \rangle$ with $s^* = F_k(r^*) \oplus m_b$.
- ③ $\mathcal{A}$ queries $\text{Dec}_k(\cdot)$ on $c' := \langle r^*, s^* \oplus \Delta \rangle$. Note $c' \neq c^*$, so the query is *allowed*.
- ④ $\mathcal{A}$ receives $m' = F_k(r^*) \oplus s^* \oplus \Delta = m_b \oplus \Delta$.
- ⑤ $\mathcal{A}$ outputs 0 if $m' \oplus \Delta = 0^n$, and 1 otherwise.

$\mathcal{A}$ wins with probability 1, since $m' \oplus \Delta = m_b$.

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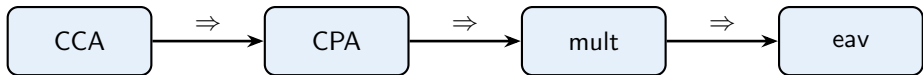
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Conclusion

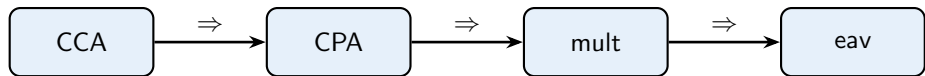
The scheme is **malleable**: flipping the last bit of s^* flips the last bit of the plaintext. A single decryption query reveals b , so Π is *not* CCA-secure. (Similar malleability attacks break every mode of operation we will see next time.)

A Hierarchy of Security Notions



- **Forward implications** are immediate: an adversary for the weaker game is an adversary for the stronger game that ignores its extra powers (except $\text{CPA} \Rightarrow \text{mult}$, which needed the hybrid argument!).

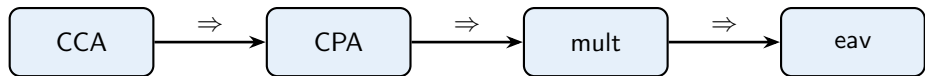
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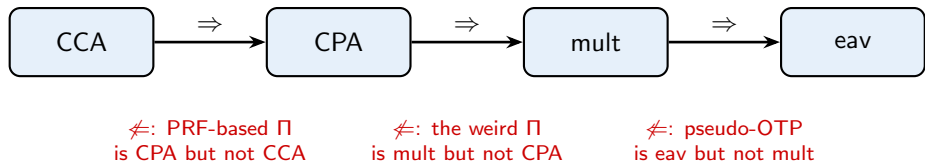


$\nRightarrow$: PRF-based Π
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$\nRightarrow$: the weird Π
is mult but not CPA

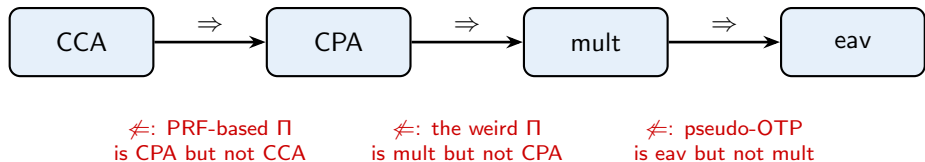
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- **Every arrow is strict**: the separating examples are the ones we have seen.

Summary

- **Mult** $\not\Rightarrow$ **CPA**: the weird scheme Π is mult-secure, but one oracle query breaks it.
- **CPA** $\Rightarrow$ **mult**: by a **hybrid argument** — some adjacent pair of the t hybrids has gap $\geq \epsilon/t$.
- **CCA**: adds a decryption oracle; the PRF-based Π is malleable, so not CCA-secure.
- **Hierarchy**: CCA $\Rightarrow$ CPA $\Rightarrow$ mult $\Rightarrow$ eav, all strict.
- **Next time**: PRPs, modes of operation, stream and block ciphers.